Reproducing Ordinary Least Squares as an Attention Mechanism

DA3410 – Assignment 11: Experiments 1–3

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October 2026

Abstract

In this report, I reproduce and evaluate three sets of experiments based on Philippe Goulet Coulombe's paper, *Ordinary Least Squares as an Attention Mechanism*. My objective is to understand how linear regression can be expressed as an attention operation, whether this representation extends to regularized estimators and how Attention Regression compares with conventional machine-learning models on simulated data.

In Experiment 1, I verify the numerical equivalence between ordinary least squares (OLS) and its attention representation. In Experiment 2, I extend the comparison to principal component regression (PCR) and ridge regression. In Experiment 3, I evaluate Attention Regression alongside OLS, random forest, gradient boosting and a multilayer perceptron (MLP).

The first two experiments establish numerical equivalence between corresponding implementations. The third experiment covers all 96 simulation conditions using one replication per condition under a limited-compute budget. Its overall model ranking agrees with the paper, although individual results differ, particularly for Attention Regression. I discuss these differences in the context of replication count, optimisation limits and model configuration.

Introduction

Ordinary least squares are a fundamental method for regression. Attention, on the other hand, is commonly associated with neural networks and sequence models. The paper *Ordinary Least Squares as an Attention Mechanism* establishes a mathematical connection between these two approaches.

The central idea is that regression predictions can be represented as weighted combinations of training targets, where the weights are computed through a data-dependent similarity operation. This allows OLS and certain regularized regression estimators to be expressed within an attention-style framework.

I organised my work into three experiments:

1. I verified that OLS and its attention representation produce numerically equivalent predictions.
2. I tested whether the same equivalence holds for PCR and ridge regression.
3. I compared Attention Regression with conventional regression and machine-learning baselines on six simulated data-generating processes.

For each experiment, I measured the relevant numerical results, summarized them in tables and compared my observations with the results and conclusions reported in the paper.

Experimental Setup

Simulation design

For the simulation experiments, I used five input features and considered six data-generating processes (DGPs):

- Linear
- Friedman 1
- Friedman 2
- Friedman 3
- Rotated Sine
- Soft Radial

The training sample sizes were

$$N \in \{500, 1000, 2500, 5000\},$$

and the signal-to-noise ratios were

$$\text{SNR} \in \{0.5, 1, 2, 3\}.$$

The test set contained 1,000 observations. Therefore, the full simulation grid contained

$$6 \times 4 \times 4 = 96$$

configurations.

I evaluated predictive performance using the out-of-sample coefficient of determination, R^2. A higher test R^2 indicates better predictive performance relative to predicting the test-set mean. A value close to zero indicates little improvement over that reference, while a negative value indicates worse performance.

Comparison methodology

I treated the paper's reported values as reference results rather than assuming that my numerical values would be identical in every experiment. This distinction is particularly important in Experiment 3, where the paper averages ten replications but my limited-compute run uses one replication per condition.

Experiment 1: OLS as an Attention Mechanism

Objective

The objective of my first experiment was to verify the paper's central mathematical result: an OLS predictor can be expressed as an attention operation without changing its predictions.

Method

I augmented the input matrix with an intercept column and computed the OLS solution. I then constructed the corresponding spectral representation using the eigenvectors and eigenvalues of the augmented design matrix's Gram matrix.

Let F_{train} and F_{test} denote the transformed training and test feature matrices. The attention weights are

$$W = F_{\text{test}} F_{\text{train}}^{\top},$$

and the predictions are

$$\widehat{y}_{\text{attention}} = W y_{\text{train}}.$$

I compared these predictions directly with conventional OLS predictions for the same observations.

Results

For the illustrative linear configuration with $N = 500$ and SNR $= 1$, I obtained a test R^2 of approximately 0.475706. The maximum absolute difference between the OLS and attention predictions was approximately

$$2.02 \times 10^{-14}.$$

Across the full benchmark, I evaluated 960 runs corresponding to 96 configurations and ten replications. The maximum prediction difference across these runs was

$$2.274 \times 10^{-13}.$$

These differences are at floating-point numerical precision. Consequently, the two implementations produced equivalent predictions for the tested configurations.

Table 1: Numerical equivalence between OLS and attention.

Measurement	**Observed value**
Illustrative test R^2	0.475706
Illustrative maximum prediction difference	2.02×10^{-14}
Full-benchmark runs	960
Full-benchmark maximum prediction difference	2.274×10^{-13}
Training-feature orthonormality error	approximately 7.55×10^{-15}
Test-feature orthonormality deviation	approximately 1.0563
Minimum attention weight	-0.017733

I also inspected the properties of the attention weights. The training features were orthonormal to numerical precision, whereas the test features were not generally orthonormal. The attention weights summed to approximately one because the intercept was included, but some weights were negative.

This observation is important: attention weights in this regression construction should not automatically be interpreted as probabilities. They can be negative while still producing the correct linear prediction.

Comparison with the paper

My results agree with the paper's central claim. The attention representation reproduced the OLS predictions to numerical precision, and this equivalence held across the complete benchmark.

The small nonzero prediction differences do not indicate a substantive disagreement. They arise from finite-precision floating-point calculations.

The weight diagnostics also support the mathematical construction: the weights need not all be positive for the attention operation to reproduce OLS. Therefore, I conclude that Experiment 1 successfully verifies the OLS-as-attention identity.

Experiment 2: Beyond OLS

Objective

After verifying the OLS identity, I investigated whether the attention representation could also reproduce two regularized regression estimators: principal component regression (PCR) and ridge regression.

The purpose was not only to compare predictive accuracy, but also to verify whether each conventional estimator and its spectral-attention counterpart produced the same predictions.

Principal component regression

PCR restricts regression to the leading principal directions of the training data. I varied the number of retained components from $L = 1$ to $L = 5$ and compared conventional PCR predictions with those obtained using the corresponding attention representation.

For an illustrative configuration, the test R^2 values for $L = 1, \ldots, 5$ were:

Table 2: Illustrative PCR test R^2 as the number of retained components increases.

Number of components L	**Test** R^2
1	-0.002212
2	0.061620
3	0.079149
4	0.233417
5	0.249118

In this example, retaining more components improved test performance. The improvement was not caused by changing from PCR to attention; it resulted from retaining more of the available feature directions.

Across the full benchmark, I performed 4,800 PCR comparisons. The maximum prediction difference between conventional PCR and its attention implementation was approximately

$$5.33 \times 10^{-14}.$$

Ridge regression

I next evaluated ridge regression using the regularization parameters

$$\lambda \in \{0.01, 0.1, 1, 10, 100\}.$$

Ridge regression modifies the inverse covariance operation by adding a regularization term. In the spectral representation, this changes the scaling of the eigen-directions while retaining the same eigenvector basis.

I compared conventional ridge regression with the corresponding regularized attention formulation. Across 4,800 comparisons, the maximum absolute prediction difference was approximately

$$6.79 \times 10^{-13}.$$

Table 3: Prediction equivalence for PCR and ridge regression.

Estimator	**Comparisons**	**Maximum prediction difference**
PCR	4,800	5.33×10^{-14}
Ridge regression	4,800	6.79×10^{-13}

Comparison with the paper

The PCR and ridge experiments agree with the paper's mathematical framework. The corresponding attention formulations reproduced the conventional predictions to numerical precision.

However, prediction equivalence does not imply that every choice of L or λ produces the same predictive accuracy. PCR can lose useful information when too few components are retained. Ridge regression can underfit when regularization is too strong.

My results therefore support two separate conclusions:

1. The attention representation reproduces the predictions of the corresponding conventional estimator.
2. Predictive performance still depends on the estimator's hyperparameters and the underlying data-generating process.

The numerical equivalence tests are particularly strong evidence because they directly compare predictions rather than relying only on aggregate R^2 values.

Experiment 3: Attention Regression and Baselines

Objective

In the third experiment, I evaluated the predictive performance of Attention Regression against four conventional baselines:

- Ordinary least squares (OLS)
- Random forest (RF)
- Multilayer perceptron (MLP)
- Gradient boosting (GBM)

I evaluated all five models across the six DGPs, four training sizes, and four SNR values. This resulted in 96 simulation conditions and 480 model evaluations for the one-replication run.

Implementation and resource limitations

I implemented Attention Regression with five attention heads, Cholesky-based initialisation, L-BFGS optimisation, and early stopping. The configuration followed the paper's general experimental design, but the available compute budget required a limited-compute run.

In particular, I used one replication per condition instead of the paper's ten replications. I also imposed a time limit on Attention Regression fits and reduced the sizes of some baseline models.

These differences are important when interpreting the results. My Experiment 3 is a limited-compute reproduction of the simulation design, not an exact numerical reproduction of the paper's full benchmark.

Overall results

The average test R^2 values across the 96 conditions were:

The overall ranking in my run was

$$\text{MLP} > \text{Attention Regression} > \text{RF} > \text{GBM} > \text{OLS}.$$

This is the same overall ranking reported in the paper. The average differences were small for all five models, particularly for random forest, gradient boosting, and Attention Regression.

Table 4: Overall test R^2: my limited-compute results versus the paper's reported averages.

Model	My result	Paper	Difference
OLS	0.288	0.279	+0.008
Random Forest	0.455	0.454	+0.001
MLP	0.530	0.521	+0.009
Gradient Boosting	0.418	0.418	+0.000
Attention Regression	0.479	0.476	+0.003

Results across data-generating processes

The average test R^2 varied across the six DGPs. This is expected because the models make different assumptions about the underlying relationship between the inputs and the target.

Table 5: My mean test R^2 by DGP and model.

DGP	OLS	RF	MLP	GBM	Attn. Reg.
Linear	0.561	0.507	0.544	0.523	0.484
Friedman 1	0.425	0.494	0.524	0.497	0.518
Friedman 2	0.045	0.373	0.522	0.287	0.482
Friedman 3	0.507	0.501	0.545	0.526	0.548
Rotated Sine	0.191	0.378	0.519	0.258	0.370
Soft Radial	-0.004	0.479	0.528	0.416	0.471

For the linear DGP, OLS obtained the highest average test R^2 among the models in my run. This is reasonable because the underlying signal is linear, making OLS a well-matched estimator.

For Friedman 2 and Soft Radial, OLS performed poorly relative to the more flexible models. These processes contain nonlinear structure that a simple linear predictor cannot fully capture. The MLP achieved strong average performance on both.

On Friedman 3, Attention Regression obtained the highest mean test R^2 in my run, although the difference from the MLP was small. For Friedman 1, Friedman 2, Rotated Sine, and Soft Radial, the MLP had the highest mean among my five models.

Where my results agree with the paper

I observed several important areas of agreement.

1. **Overall ranking:** the MLP ranked first, followed by Attention Regression, random forest, gradient boosting, and OLS. This matches the overall ranking reported in the paper.
2. **Linear data:** OLS achieved the highest mean test R^2 in my linear-DGP results. This is consistent with the advantage of using a correctly specified linear estimator when the true relationship is linear.
3. **Nonlinear data:** flexible models generally performed better than OLS on the more challenging nonlinear processes. In particular, OLS was weak on Friedman 2, Rotated Sine, and Soft Radial.
4. **Overall Attention Regression performance:** its mean test R^2 was 0.479, compared with 0.476 in the paper. The difference is approximately 0.003.

These findings indicate that my limited-compute results reproduce the broad model-performance patterns described in the paper.

Where my results differ from the paper

Although the overall ranking agreed, the best model for an individual DGP did not always match the paper.

Table 6: Best model by DGP: my results compared with the paper.

DGP	My best model	Paper's best model
Linear	OLS	OLS
Friedman 1	MLP	Attention Regression
Friedman 2	MLP	MLP
Friedman 3	Attention Regression	MLP
Rotated Sine	MLP	MLP
Soft Radial	MLP	MLP

The clearest differences occurred for Friedman 1 and Friedman 3. My MLP performed best on Friedman 1, whereas the paper reported Attention Regression as the best model. On Friedman 3, my Attention Regression result was slightly higher than the MLP, whereas the paper reported the MLP as best.

I also observed some larger condition-specific differences for Attention Regression. For example, on Rotated Sine with $N = 1000$ and SNR $= 3$, my test R^2 was approximately 0.17, compared with 0.56 in the paper. On the linear DGP with $N = 5000$ and SNR $= 3$, my result was approximately 0.42, compared with 0.57 in the paper.

Table 7: Examples of condition-specific differences in Attention Regression test R^2.

Condition	My result	Paper	Difference
Rotated Sine, $N = 1000$, SNR $= 3$	0.17	0.56	-0.39
Linear, $N = 5000$, SNR $= 3$	0.42	0.57	-0.15
Linear, $N = 5000$, SNR $= 2$	0.37	0.52	-0.15
Friedman 1, $N = 5000$, SNR $= 3$	0.58	0.72	-0.14
Linear, $N = 2500$, SNR $= 1$	0.29	0.43	-0.14

These differences show why an overall mean alone is not sufficient to establish an exact replication.

Explaining the differences

I attribute the differences primarily to the limitations of the third experiment's setup.

First, I used one replication per condition, while the paper averaged ten replications. A single simulation can be affected by the particular training and test samples generated, so its R^2 may differ from the ten-replication average.

Second, the limited-compute implementation capped the optimisation time for Attention Regression and reduced the sizes of some baseline models. A time-capped optimisation may stop before reaching the solution obtained under the paper's full training procedure.

Third, differences in individual DGPs can arise because each model has a different inductive bias and optimisation behaviour. A model that performs best on average need not be the best model for every individual configuration.

I therefore interpret the close overall averages as evidence that my implementation captures the broad patterns, but I do not treat the one-replication results as evidence that every condition has been reproduced numerically.

Overall Discussion

The three experiments provide complementary evidence about the relationship between regression and attention.

In Experiment 1, I verified that the OLS attention construction reproduces conventional OLS predictions to numerical precision. This was a direct numerical test of the paper's mathematical identity.

In Experiment 2, I extended this verification to PCR and ridge regression. The results showed that spectral-attention representations can reproduce these estimators as well. However, the choice of components or regularization parameter still influences predictive accuracy.

In Experiment 3, I compared Attention Regression with four baseline models. The overall ranking matched the paper, and the overall average R^2 for Attention Regression was close to the reference result. However, individual DGP results differed, which is unsurprising given the limited replication count and optimisation budget.

The distinction between these findings is important. Experiments 1 and 2 test *numerical equivalence*: two implementations should generate the same predictions. Experiment 3 tests *predictive performance*: different models are trained and compared, so their results depend on the data, model configuration, and optimisation procedure.

Limitations

I considered the following limitations when interpreting the results:

- Experiment 3 used one replication per condition rather than the ten replications used in the paper.
- The attention optimisation time limit and reduced baseline model sizes may affect predictive performance.
- The reported overall averages should not be interpreted as a substitute for replication-level uncertainty estimates.
- The numerical comparisons are based on the tested configurations and do not establish that all possible implementations will behave identically.

Experiments 1 and 2 showed extremely small prediction differences between mathematically equivalent implementations. Experiment 3, in contrast, was designed as a limited-compute comparison and should be interpreted accordingly.

Conclusion

In this work, I investigated the representation of regression estimators as attention operations and evaluated Attention Regression against conventional machine-learning baselines.

My first experiment verified the OLS-as-attention identity, with a maximum prediction difference of approximately 2.274×10^{-13} across the full benchmark. My second experiment extended the same numerical verification to PCR and ridge regression, with maximum differences of approximately 5.33×10^{-14} and 6.79×10^{-13}, respectively.

My third experiment evaluated five models across all 96 simulation conditions. The overall ranking matched the paper, and the average test R^2 for Attention Regression was 0.479, compared with 0.476 in the paper. Some individual conditions and the best-performing models for two DGPs differed.

I conclude that Experiments 1 and 2 successfully verified the numerical equivalence of the corresponding estimators and their attention representations. Experiment 3 reproduced the broad performance patterns

of the paper, but the limited replication count and optimisation budget mean that it should be viewed as a preliminary, resource-constrained reproduction rather than an exact numerical replication.